

Phase 1: EDA Report - Rebuilding the Social Engine

1. Data Profiling Summary

Raw Data Overview

Dataset	Rows	Columns	Duplicates
Posts (Corrupted)	12,360	8	360
Users	1,500	5	0

Column-Level Quality Assessment (Posts)

Column	True Nulls	'NULL' Strings	Unique Values	Issue Type
post_id	0	0	12,000	360 duplicate rows
user_id	0	0	1,500	All valid FK references
platform	1,846	0	5	~15% missing platform
text_content	1,746	24	10,225	HTML artifacts, encoding errors
timestamp	0	0	8,839	3 mixed formats

likes	1,858	0	4,752	525 negative values
shares	0	0	1,994	Clean
comments	0	0	1,001	Clean

Users Table

- **1,500 unique users** across **33 locations** and **10 languages**
 - Account creation dates span all of 2023
 - Follower counts range from ~100 to ~50,000
 - **No missing values, no duplicates** - clean reference data
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2. Data Cleaning Transformations & Justifications

2.1 Duplicate Row Removal (360 rows)

Action: Dropped 360 exact duplicate rows from Posts table (kept first occurrence). **Justification:** The corrupted dataset contained 360 fully duplicated rows (identical across all 8 columns). These represent data ingestion errors, not legitimate reposts. Keeping them would inflate engagement metrics.

2.2 'NULL' String Replacement (24 in text_content)

Action: Converted literal "NULL" strings to proper NaN. **Justification:** The corrupted export used the string "NULL" as a sentinel value instead of proper null markers. 24 occurrences in text_content were converted to maintain consistent null semantics across the pipeline.

2.3 HTML & Encoding Artifact Removal

Action: Stripped `<br>`, `<div>`, `</div>` HTML tags, decoded `&` entities, removed `Ã©` encoding artifacts. **Justification:** The data contains injected HTML tags and UTF-8 mojibake (`Ã©` = mis-encoded `é`). These are data corruption tokens from a faulty export pipeline, not genuine user content. Removal preserves the actual text semantics.

2.4 Timestamp Format Unification

Action: Parsed three distinct timestamp formats into unified `datetime`:

- ISO 8601: `2024-05-05T05:52:34`
- DD-MM-YYYY: `25-09-2024`
- Unix epoch: `1722528840`

Justification: The system stored timestamps in multiple formats due to schema migration or multi-source ingestion. Unifying to `datetime` enables temporal analysis. Date range: **2024-05-01 to 2025-04-30** (12 months of data).

2.5 Negative Engagement Values (525 in likes)

Action: Took absolute value of 525 negative like counts. **Justification:** Social media engagement metrics (likes, shares, comments) are inherently non-negative. Negative values (e.g., -4812, -3707) represent sign-corruption during data transfer. The absolute value recovers the most likely intended magnitude.

2.6 Platform Name Standardization

Action: Title-cased all platform names, normalized "Youtube" to "YouTube". **Justification:** Ensures consistent grouping. Missing platforms (14.9%) are preserved as NaN since we cannot impute platform from content alone.

2.7 Feature Engineering

New derived columns created for downstream analysis:

- `hashtags`: Extracted from `text_content` using regex
- `brand_mentioned`: Identified from 10 known brands in the dataset
- `sentiment`: Rule-based classification (positive/negative/neutral) from keyword patterns
- `total_engagement`: likes + shares + comments
- `engagement_rate`: `total_engagement` / 3
- `post_date`, `post_hour`, `post_day_of_week`, `post_month`: Temporal decomposition

3. Core EDA Insights

3.1 Engagement Statistics

Metric	Mean	Median	Std Dev	Min	Max
Likes	2,499	2,530	1,433	1	5,000
Shares	987	990	574	0	2,000
Comments	491	494	286	0	999
Total Engagement	3,631	3,640	1,557	17	7,920

[!NOTE] Engagement follows a near-uniform distribution within bounded ranges, suggesting synthetic data generation with controlled parameters. Likes dominate total engagement (~69%).

3.2 Platform Dynamics

Platform	Posts	% Share	Avg Engagement	Unique Users
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YouTube	2,136	17.8%	3,607	1,147
Facebook	2,135	17.8%	3,660	1,173
Twitter	2,119	17.7%	3,563	1,132
Reddit	2,086	17.4%	3,639	1,132
Instagram	2,038	17.0%	3,650	1,139
Missing	1,784	14.9%	3,644	1,063

Insight: Posts are nearly uniformly distributed across all 5 platforms. Facebook and Instagram lead slightly in average engagement, but the difference is marginal (~3%). This suggests a platform-agnostic user base.

3.3 Temporal Patterns

Monthly Volume: Consistent posting volume (~920-1,010 posts/month) across the 12-month window, with no seasonal decline, indicating healthy platform retention.

Day-of-Week: Activity is evenly distributed across all 7 days (~1,700 posts/day), with no significant weekday vs. weekend skew.

Peak Activity Hours: Engagement peaks show no strong hourly pattern, with posts distributed across all 24 hours.

Spike Days Detected (>2 standard deviations above mean):

Date	Posts	Significance
2024-06-16	54	Highest single-day volume
2024-05-30	51	Early spike
2024-08-14	50	Mid-summer surge
2025-01-25	49	New Year period
2025-02-19	48	February activity

3.4 Brand Mention Analysis

Brand	Mentions	Avg Likes	Avg Shares	Avg Engagement	Unique Users
Nike	1,345	2,507	969	3,645	996
Google	1,334	2,498	979	3,602	990
Samsung	1,291	2,483	1,003	3,635	964
Amazon	1,279	2,535	971	3,612	966
Apple	1,268	2,519	977	3,652	960
Pepsi	1,261	2,486	1,006	3,664	956
Microsoft	1,241	2,479	991	3,596	939
Toyota	1,226	2,460	999	3,609	947
Coca-Cola	1,221	2,540	996	3,660	936
Adidas	1,197	2,486	1,005	3,626	931

Insight: Nike dominates brand mentions (1,345), but Coca-Cola and Pepsi lead in average engagement. All 10 brands have remarkably similar engagement profiles, suggesting uniform audience interest across brand categories.

3.5 Sentiment Distribution

Sentiment	Posts	Avg Engagement
Positive	3,651	3,635
Negative	3,494	3,611
Neutral	4,855	3,643

Insight: Sentiment is roughly balanced with a slight positive skew. Engagement rates are virtually identical across sentiments, meaning negative content is NOT penalized by the engagement algorithm - a

potential concern for content moderation.

3.6 User Segmentation

Segment	Count	Description
Regular User	1,010	4-14 posts, moderate engagement
High Engager	419	Avg engagement > 4,000
Casual User	46	1-3 posts total
Power User	25	Top 1% by post volume (15+ posts)

- **Average posts per user:** 8.0 (median: 8.0)
- **Power users** (25 users) have avg engagement of 3,612 - in line with the population mean
- **Zero bot-like accounts** detected (no user has high frequency + low engagement)

3.7 Geographic Distribution

Top cities by post volume: Los Angeles (459), Munich (452), Shanghai (451), Barcelona (439) **Highest engagement cities:** London (3,702), Osaka (3,700), Barcelona (3,692), Los Angeles (3,685)

Language engagement ranking: Arabic (ar) leads at 3,696 avg engagement, followed by Chinese (zh) at 3,677. Hindi trails at 3,497.

3.8 Correlation Analysis

	Likes	Shares	Comments
Likes	1.000	-0.001	0.010
Shares	-0.001	1.000	0.024
Comments	0.010	0.024	1.000

Key Finding: Likes, shares, and comments are **completely**

uncorrelated with each other (correlations near zero). This is unusual for real social media where these metrics typically co-vary. Follower count shows **no correlation** with engagement (-0.011), suggesting engagement is organic and not follower-driven.

3.9 Content Completeness

Field	Completeness
text_content	85.3%
platform	85.1%
likes	84.9%
shares	100%
comments	100%

- **239 posts** are missing BOTH text content and platform metadata simultaneously

4. Data Quality Assessment Summary

[!IMPORTANT] The cleaned dataset contains **12,000 posts** from **1,500 users** spanning **May 2024 to April 2025**. After removing 360 duplicates, fixing 525 negative values, converting 24 NULL strings, and stripping HTML artifacts, the dataset is production-ready for Phase 2 SQL ingestion.

Remaining Missing Data (by design - not imputable)

- **Platform** (14.9%): Cannot reliably impute posting platform from content
- **Text content** (14.3%): Posts may have been deleted or corrupted beyond recovery
- **Likes** (15.1%): Engagement data may not have been captured for

all posts

- **Brand** (14.8%): Not all posts mention a brand - this is expected